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Acronyms

AI	Artificial Intelligence
API	Application Programming Interface
ATMs	Automatic Teller Machines
CAWI	Computer-Assisted Web Interview
CEDLAS	Center for Distributive, Labor and Social Studies
CEE	Central and Eastern European
CEPS	Centre for European Policy Studies
GBB	Gmyrek, Berg and Bescond, 2023. ILO Working Paper 96
GPT-4	Generative Pre-trained Transformer 4
GPT-4o	Generative Pre-trained Transformer 4 Omni
GUS	Główny Urząd Statystyczny (Central Statistical Office, Poland)
HIC	High-Income Countries
ICT	Information and Communication Technology
ILO	International Labour Organization
IMF	International Monetary Fund
ISCO-08	International Standard Classification of Occupations (2008 version)
IT	Information Technology
ITC-ILO	International Training Centre of the ILO (Turin)
LFS	Labour Force Survey (BAEL in Poland)
LICs	Low-Income Countries
LLM	Large Language Model
ML	Machine Learning
MOL	Ministry of Family, Labour and Social Policy in Poland
NASK-PIB	National Research Institute

O*NET	Occupational Information Network (U.S. Department of Labor)
PIE	Polski Instytut Ekonomiczny (Polish Economic Institute)
SD	Standard Deviation
SGH	Szkoła Główna Handlowa (Warsaw School of Economics)
U.S. / US / USA	United States of America
UK	United Kingdom
UNLP	National University of La Plata

► Introduction

In August 2023, the ILO published the first global employment estimates of potential occupational exposure to generative artificial intelligence (GenAI) (Gmyrek et al., 2023). This research was prompted by concerns – that have featured prominently in the media – of the potential for the technology to replace large shares of knowledge work.

Since the release of the 2023 estimates, there have been further technological advances in GenAI, including the introduction of Large Language Models (LLMs) with expanded capabilities in image recognition and audio and video production. Moreover, many new GenAI applications have been developed as a result of the possibility of linking GenAI models with Application Programming Interface (API) tools. Indeed, GenAI has been likened to a general-purpose technology, meaning that it may continue generating new applications and innovations across various sectors, driving long-term economic and societal transformations.

Despite this rapidly developing landscape, it is important for policy makers to develop deeper understanding of how technological advances might impact existing jobs. To address this need, the Research department of the ILO has partnered with NASK, the National Research Institute of the Ministry of Digital Affairs in Poland, to develop a more precise assessment of potential effects of GenAI on employment across countries.

This study builds on the method developed by the ILO (Gmyrek, Berg and Bescond, 2023 - GBB hereafter), which used task descriptions from the 4-digit International Standard Classification of Occupations (ISCO-08) and the GPT-4 model to estimate task-level and occupational scores of exposure to GenAI technology and then derives global employment estimates based on official ILO labour force data. But the study includes several key refinements.

The first step in our analysis is based on Poland's national 6-digit classification of occupations that includes nearly 30,000 tasks. This expands tenfold the number of tasks given in the ISCO-08 structure and enables a fine-grained assessment of tasks' automation potential, based on a more up-to-date set of occupations and tasks than those captured in the ISCO-08 system. As Poland's 6-digit classification is aligned with the 4-digit ISCO-08, the occupational scores can be compiled at the 4-digit level allowing a straightforward comparison and updating of the 2023 estimates of the ILO (GBB). In addition, the study improves the methodological framework of GBB, by drawing on the combined strengths of human and AI abilities, with an additional layer of expert verification.

Specifically, we surveyed 1,640 people currently employed in Poland in each ISCO-08 1-digit group to rank the automation potential of a representative sample of tasks that pertain to their occupational group. As the second step, we asked a mix of national and international experts to conduct a detailed review of a large sample of tasks previously assessed in the survey to validate or adjust the level of automation scores through an iterative process. Since we maintain the hierarchical link to ISCO-08 and focus on technological feasibility rather than specific country context, by combining opinions of local and international experts, we create a system that enables more precise predictions of the potential of automation of individual occupational tasks with GenAI technology, including at the international level. To achieve that, we input the human scores into an AI model, to generate scores reflecting human judgement for nearly 30,000 individual tasks pertaining to 2,500 occupations at 6-digit level in Poland. We then use the same AI predictor to re-generate scores for the tasks in ISCO-08, previously provided by GBB (2023), thereby adjusting the 2023 index of the ILO, which now benefits from both the input of people